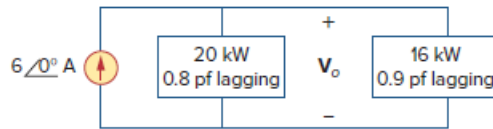


Problem

For the circuit in Fig. 11.79, find $\mathbf{V}_o$ and the input power factor.

Figure 11.79:



Step-by-step solution

Step 1 of 5

Refer to Figure 11.79 in the textbook.

Calculate the power factor angle of load 1.

$$\cos \theta_1 = 0.8 \text{ (lagging)}$$

$$\begin{aligned} \theta_1 &= \cos^{-1}(0.8) \\ &= 36.87^\circ \end{aligned}$$

Calculate the apparent power of load 1.

$$\begin{aligned} \mathbf{S}_1 &= \frac{P_1}{\cos \theta_1} \angle \cos^{-1} \theta_1 \\ &= \frac{20 \times 10^3}{0.8} \angle \cos^{-1} 0.8 \\ &= 25 \angle 36.87^\circ \text{ kVA} \end{aligned}$$

Step 2 of 5

Calculate the power factor angle of load 2.

$$\cos \theta_2 = 0.9 \text{ (lagging)}$$

$$\begin{aligned} \theta_2 &= \cos^{-1}(0.9) \\ &= 25.84^\circ \end{aligned}$$

Calculate the apparent power of load 2.

$$\begin{aligned} \mathbf{S}_2 &= \frac{P_2}{\cos \theta_2} \angle \cos^{-1} \theta_2 \\ &= \frac{16 \times 10^3}{0.9} \angle \cos^{-1} 0.9 \\ &= 17.78 \angle 25.84^\circ \text{ kVA} \end{aligned}$$

Step 3 of 5

Calculate the total apparent power of the load.

$$\begin{aligned} \mathbf{S}_T &= \mathbf{S}_1 + \mathbf{S}_2 \\ &= (25 \angle 36.87^\circ \text{ kVA}) + (17.78 \angle 25.84^\circ \text{ kVA}) \\ &= 42.59 \angle 32.29^\circ \text{ kVA} \end{aligned}$$

Step 4 of 5

Determine the value of voltage, V_o .

$$V_o = \frac{S_t}{I^*}$$

Substitute $42.59 \angle 32.29^\circ$ kVA for S_t and $6 \angle 0^\circ$ A for I in the equation.

$$\begin{aligned} V_o &= \frac{42.59 \times 10^3 \angle 32.29^\circ}{(6 \angle 0^\circ)^*} \\ &= 7.098 \angle 32.29^\circ \text{ kV} \end{aligned}$$

Thus, the value of voltage, V_o is $7.098 \angle 32.29^\circ$ kV.

Step 5 of 5

Calculate the input power factor.

$$\begin{aligned} \text{pf} &= \cos(\theta_v - \theta_i) \\ &= \cos(32.29^\circ - 0^\circ) \\ &= 0.8454 \text{ (lagging)} \end{aligned}$$

Thus, the input power factor is 0.8454 (lagging).